Ecological Evaluation of Sunder Nursery

Using the Contingent Valuation Method

RDL725 - Field Assignment

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Introduction: Why Value the Environment?

The Problem:

- Urban parks and heritage sites provide enormous benefits: recreation, ecological balance, cultural preservation, aesthetic pleasure
- These benefits have **no market price**
- Policy-makers tend to undervalue what they cannot price

Our Objective:

- Estimate the **Total Economic Value (TEV)** of Sunder Nursery using CVM and TCM
- Capture both **use values** (recreation) and **non-use values** (bequest, existence)

Field Visit Details

- **Site:** Sunder Nursery, New Delhi
- **Date:** Friday, 10 April 2026
- **Time:** 4:00 PM - 7:30 PM
- **Sample:** 45 visitors surveyed
- **Method:** Bilingual Google Forms (English-Hindi)

About Sunder Nursery

History & Restoration:

- 90-acre heritage park in Nizamuddin, adjacent to Humayun's Tomb (UNESCO)
- Origins in the 16th century as a Mughal-era nursery
- Restored by **Aga Khan Trust for Culture** (2007-2018)
- Reopened in **February 2018**

Ecological Features:

- 300+ tree species, 80+ flowering varieties
- 80+ bird species recorded
- 6 restored Mughal-era monuments
- Restored water channels & wetlands

Garden of Delight at dusk

Theoretical Framework

Total Economic Value (TEV):

$$\text{TEV} = \underbrace{\text{Use Value}}_{\text{direct benefit}} + \underbrace{\text{Non-Use Value}}_{\text{option + bequest + existence}}$$

- **Use Value:** Walking, picnicking, birdwatching
- **Option Value:** Keeping the option to visit in the future
- **Bequest Value:** Preserving for future generations
- **Existence Value:** Knowing the park exists and is protected

Contingent Valuation Method (CVM)

Stated-preference technique:
Creates a hypothetical market via survey; asks WTP for conservation

Payment vehicle: *Increased entry fee*

Travel Cost Method (TCM)

Revealed-preference technique:
Infers value from actual travel expenditure

Consumer surplus = benefit above actual spend

Survey Design & Data Collection

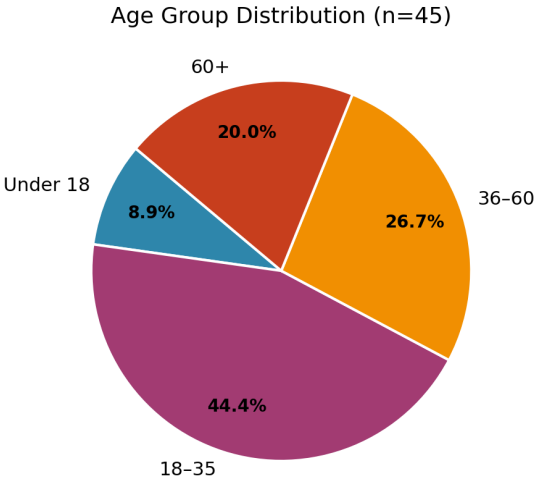
Questionnaire Structure (16 questions):

1. **CVM Core:** Aspects valued + WTP (Yes/No)
2. **Contribution:** Max extra amount (Rs. 10/20/50/100)
3. **Protest Feedback:** Why not willing
4. **Non-Use Values:** Bequest importance + existence value + ecological rating
5. **TCM Core:** Transport mode, travel cost, travel time, sole destination
6. **Demographics:** Frequency, age, income

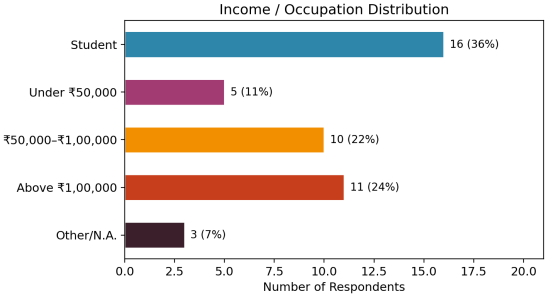
⋆ **Key Insight:** Skip logic directed WTP=Yes to amount page, WTP=No to protest page.

Conducting the survey on-site

Respondent Demographics (n = 45)

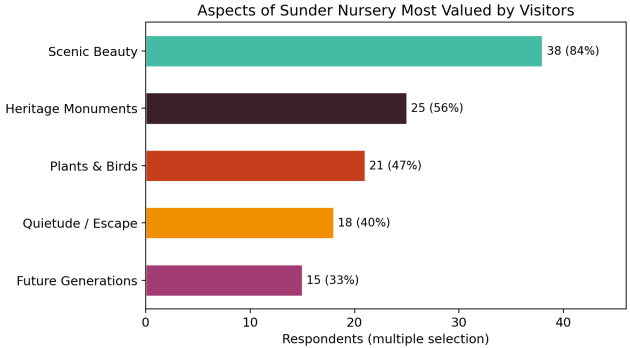


Age group distribution



Income/occupation distribution

What Do Visitors Value Most?



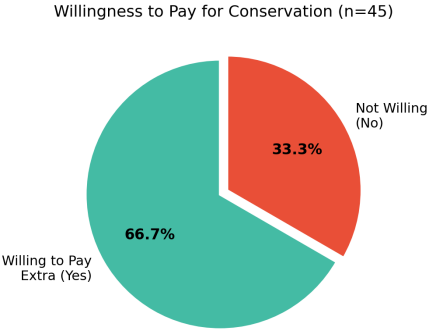
Aspects of Sunder Nursery most valued

Top aspects (multiple selections):

1. Scenic beauty: **84.4%**
2. Heritage monuments: **55.6%**
3. Plants & birds: **46.7%**
4. Quietude/escape: **40.0%**
5. Future generations: **33.3%**

⋆ **Key Insight:** A third of visitors explicitly valued "availability for future generations" - a strong signal of latent bequest value.

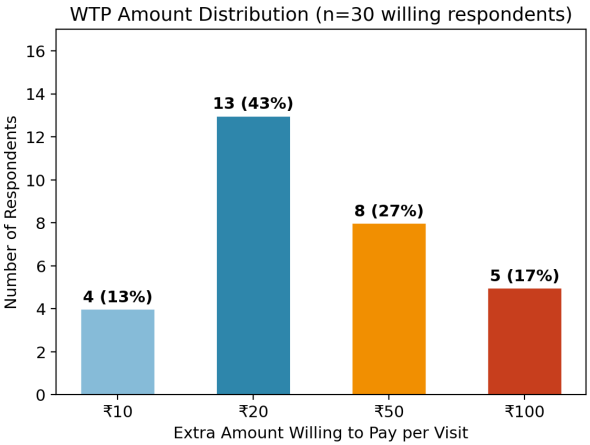
CVM Results: Willingness to Pay



WTP for higher entry fee

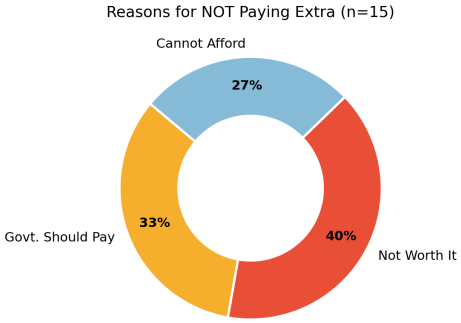
66.7% said **Yes**

33.3% said No



Max extra amount willing to pay

Protest Bid Analysis



Reasons for not paying extra

Among 15 "No" respondents:

- 40% - Not worth paying more
- 33% - Govt. should bear the cost
- 27% - Cannot afford to pay more

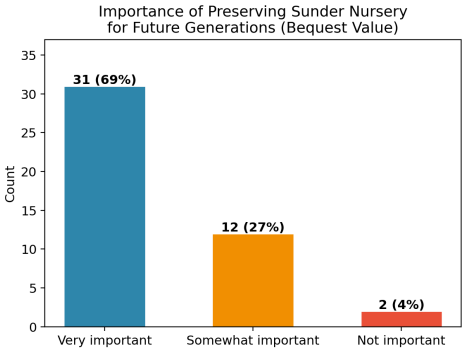
Protest Bids

The "government should pay" responses (5 respondents) are **protest bids** - they object to the payment vehicle, not the park's value.

Excluding protest bids:
Adjusted WTP rate $= \frac{30}{40} =$ **75%**

Non-Use Value Analysis

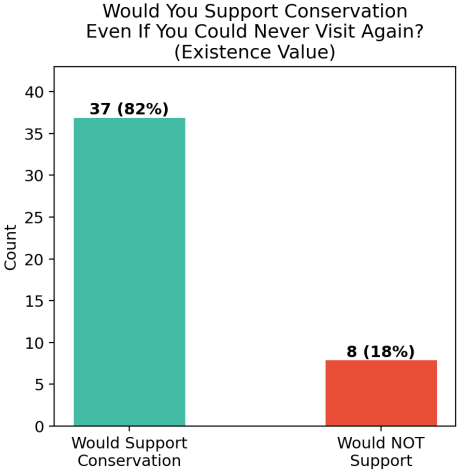
Bequest Value:



Importance of preservation for future generations

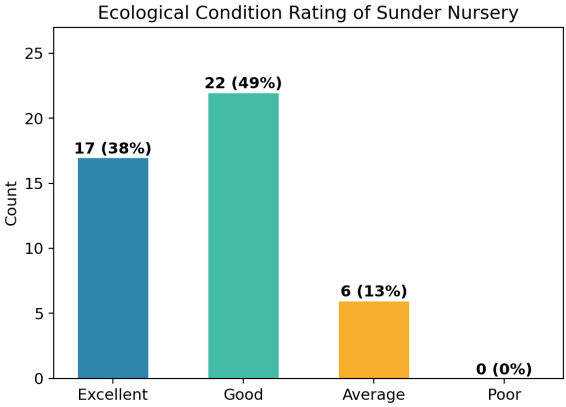
68.9% said "Very important"
26.7% "Somewhat important"

Existence Value:



Support conservation even if never visiting again?

Ecological Condition Rating



Ecological condition rating by visitors

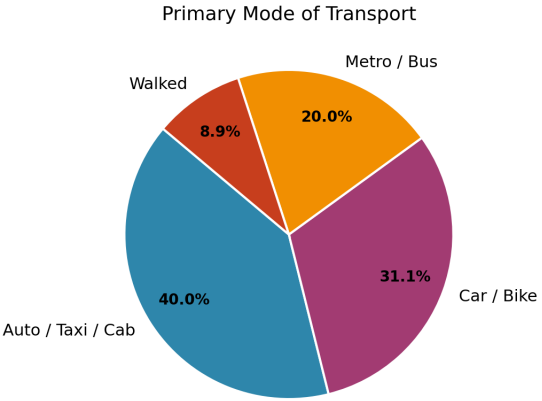
Visitor perception:

- Excellent: **37.8%**
- Good: **48.9%**
- Average: 13.3%
- Poor: 0%

Validation

86.7% rate it "Good" or "Excellent" - validating the Aga Khan Trust restoration program's ecological success.

TCM Results: Travel Cost & Consumer Surplus

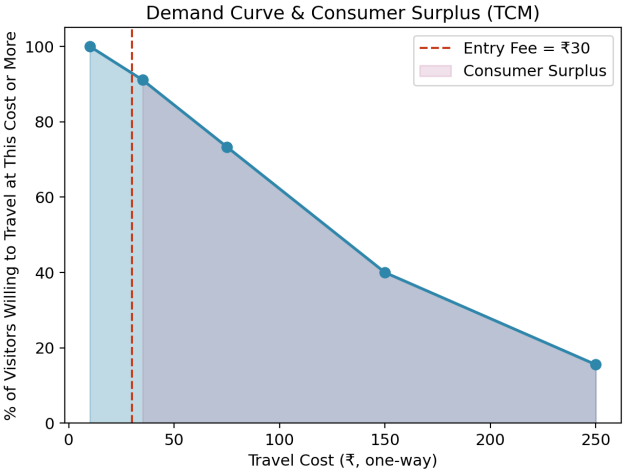


Transport mode distribution

Statistic	Value
Mean one-way cost	Rs. 107.7
Mean round-trip	Rs. 215.3
CS per visitor	**Rs. 142.3**
Est. annual visitors	∼15 lakh
Annual CS	**Rs. 21.35 cr**

- Auto/taxi most common (40%)
- 53.3% also visiting nearby sites (Humayun's Tomb)
- 51.1% travel 31-60 min one-way

Demand Curve & Consumer Surplus



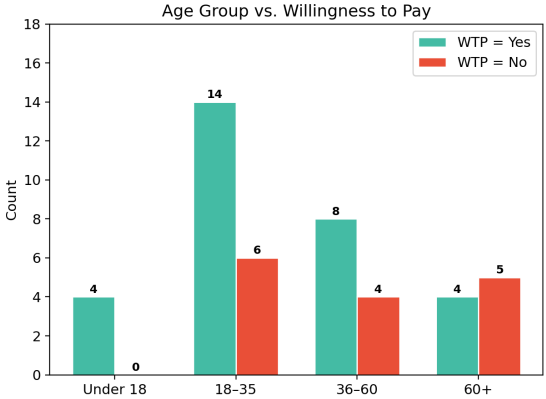
Demand curve estimated via TCM

Interpretation:

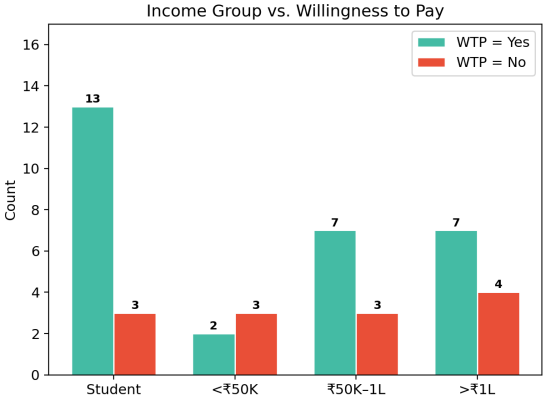
- Downward-sloping curve confirms theory: higher travel cost $\rightarrow$ fewer visitors
- Shaded area $=$ **consumer surplus** (benefit above actual spend)
- Current entry fee ($\mathrm{Rs.}$ 20-30) captures only a tiny fraction of the true value

$\star$ **Key Insight:** Visitors reveal their true valuation through what they spend to get there, not just the entry fee.

Cross-Tabulations: Who is Willing to Pay?



Age group vs. WTP



Income group vs. WTP

Under 18: 100% WTP → 60+: 44% WTP

Students: highest WTP (81%)

→ **Key Insight:** Environmental concern can outweigh income effects – students show the highest WTP rate

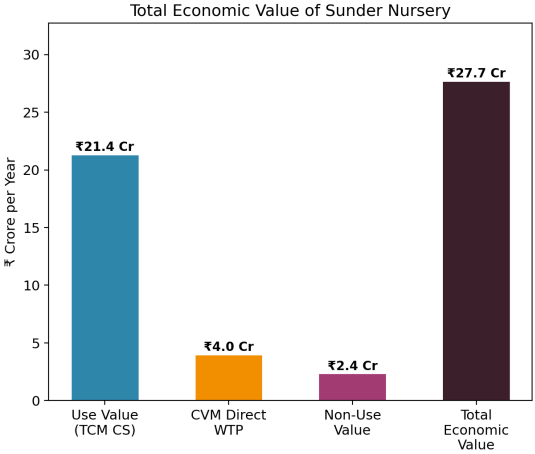
Total Economic Value of Sunder Nursery

Component	Method	Rs. cr/yr
Use Value (CS)	TCM	21.35
Direct WTP	CVM	4.00
Non-Use Value	CVM $\times$ 0.59	2.36
Total		**27.71**

Non-use multiplier:

$$1 + (0.689 \times 0.5) + (0.822 \times 0.3) = 1.59$$

Non-use component: $(1.59 - 1) \times \text{Rs. } 4.00 = \text{Rs. } 2.36$ cr



TEV breakdown

Discussion: Comparison & Policy Implications

Comparison with Literature:

- **Lumpinee Park, Bangkok** [1]: Non-use values increased capitalized value by $\sim 9\times$; our multiplier ($1.59\times$) is more conservative
- **Colorado Wilderness** [4]: Non-use = 37% of total; ours is 8.5%, reasonable given large TCM use value
- Survey design mirrors **Kalamazoo Nature Center** [7]

Policy Implications:

1. **Entry fee revision:** A modest Rs. 20-50 increase acceptable to majority; could generate Rs. 3-7.5 cr/yr extra
2. **Public investment:** 33% protest bidders want govt. funding - supports continued public investment
3. **Youth engagement:** High student WTP suggests outreach programs would strengthen conservation support

Limitations

CVM Biases [5]

1. **Strategic bias:** Free-rider behavior or "warm glow" overstating
2. **Hypothetical bias:** No real money changed hands
3. **Information bias:** Limited info in a 2-min survey

Sampling Limitations

4. **Sample bias:** Single Friday evening; weekday/morning visitors may differ
5. **Small sample:** $n = 45$; larger surveys (200+) needed for robust policy

⋆ **Key Insight:** Despite limitations, results are directionally consistent with established literature.

Conclusion

Key Findings

1. **Strong WTP:** 66.7% willing to pay higher entry fee; mean WTP $= \mathrm{Rs.}\,40$/visit
2. **Robust non-use values:** 68.9% bequest ("very important") + 82.2% existence value
3. **Ecological success:** 86.7% rate condition as Good/Excellent
4. **Significant consumer surplus:** $\mathrm{Rs.}\,142$/visitor $\rightarrow$ $\mathrm{Rs.}\,21.35$ crore/year

Total Economic Value $\approx$ $\mathrm{Rs.}\,27.71$ crore/year

Sunder Nursery's value is largely invisible to the market, but is very much perceived and cherished by those who visit it - and even by those who simply want to know it exists.

References

[1] Dixon, J.A. and Hufschmidt, M.M. (1986). *Economic Valuation Techniques for the Environment*. Johns Hopkins University Press.

[2] Krutilla, J.V. (1967). Conservation reconsidered. *American Economic Review*, 57(4).

[3] Johansson, P.O. (1990). *The Economic Theory and Measurement of Environmental Benefits*. Cambridge UP.

[4] Walsh, R.G., Loomis, J.B., and Gillman, R.A. (1984). Valuing option, existence, and bequest demands for wilderness. *Land Economics*, 60(1).

[5] Schulze, W.D., d'Arge, R.C., and Brookshire, D.S. (1981). Valuing environmental commodities. *Western J. Agric. Econ.*, 6(2).

[6] Aga Khan Trust for Culture (2018). *Sunder Nursery Heritage Park*. New Delhi.

[7] RDL725 Course Material (Prof. Vivek Kumar, IIT Delhi).

Thank You

Open for Questions

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